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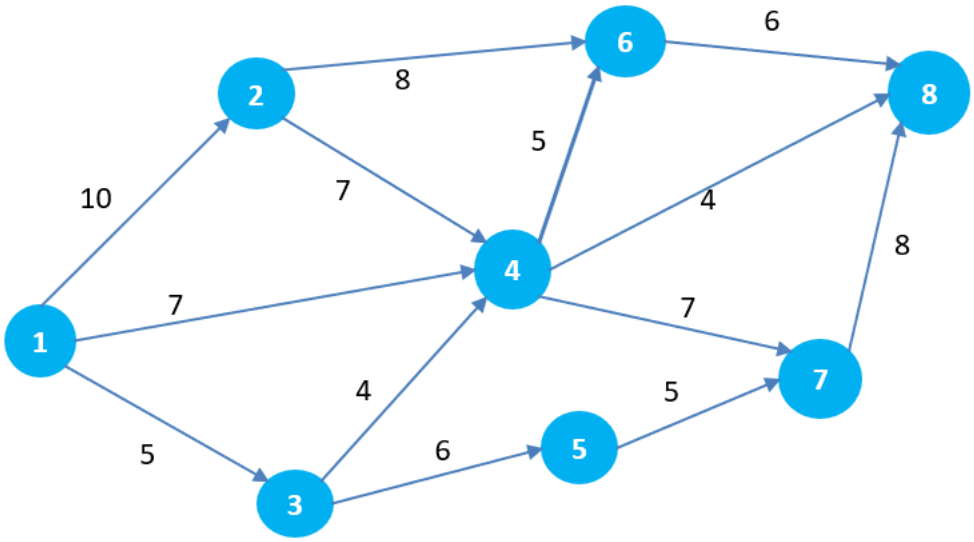
AY 2025 E211 Operations Planning II
FA Workshop Exercises Solution

Question 1

Determine the shortest path from node 1 to each of the other nodes using Dijkstra's algorithm. The values on the arcs represent the distance between 2 nodes.



Iteration	Solution Set	Arcs	Total Distance
Iteration 1	1	1 → 2 1 → 3 1 → 4	10 5 7
Iteration 2	1 3 (5)	1 → 2 1 → 4 3 → 4 3 → 5	10 7 9 11
Iteration 3	1 3 (5) 4 (7)	1 → 2 3 → 5 4 → 6 4 → 7 4 → 8	10 11 12 14 11
Iteration 4	1 3 (5) 4 (7) 2 (10)	3 → 5 4 → 6 4 → 7 4 → 8 2 → 6	11 12 14 11 18

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Iteration 5	1 3 (5) 4 (7) 2 (10) 5 (11)	4 → 6 4 → 7 4 → 8 2 → 6 5 → 7	12 14 11 18 16
Iteration 6	1 3 (5) 4 (7) 2 (10) 5 (11) 8 (11)	4 → 6 4 → 7 2 → 6 5 → 7	12 14 18 16
Iteration 7	1 3 (5) 4 (7) 2 (10) 5 (11) 8 (11) 6 (12)	4 → 7 5 → 7	14 16
Iteration 8	1 3 (5) 4 (7) 2 (10) 5 (11) 8 (11) 6 (12) 7 (14)		

From 1 to	Shortest Path	Total Distance
2	1→2	10
3	1→3	5
4	1→4	7
5	1→3→5	11
6	1→4→6	12
7	1→4→7	14
8	1→4→8	11

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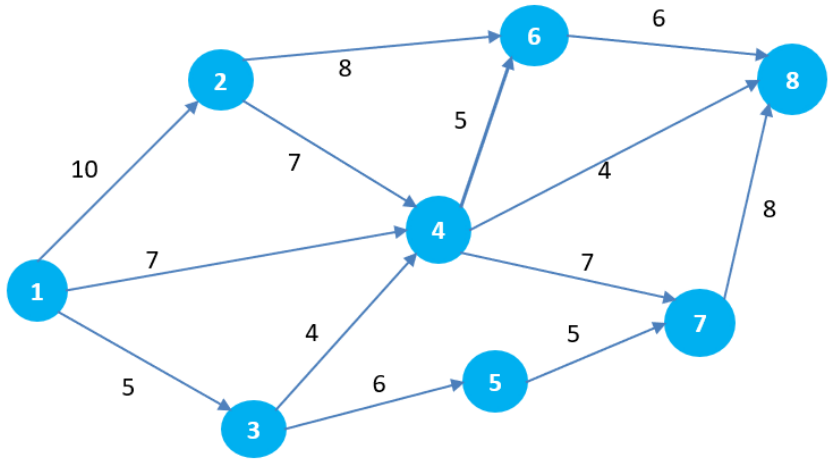
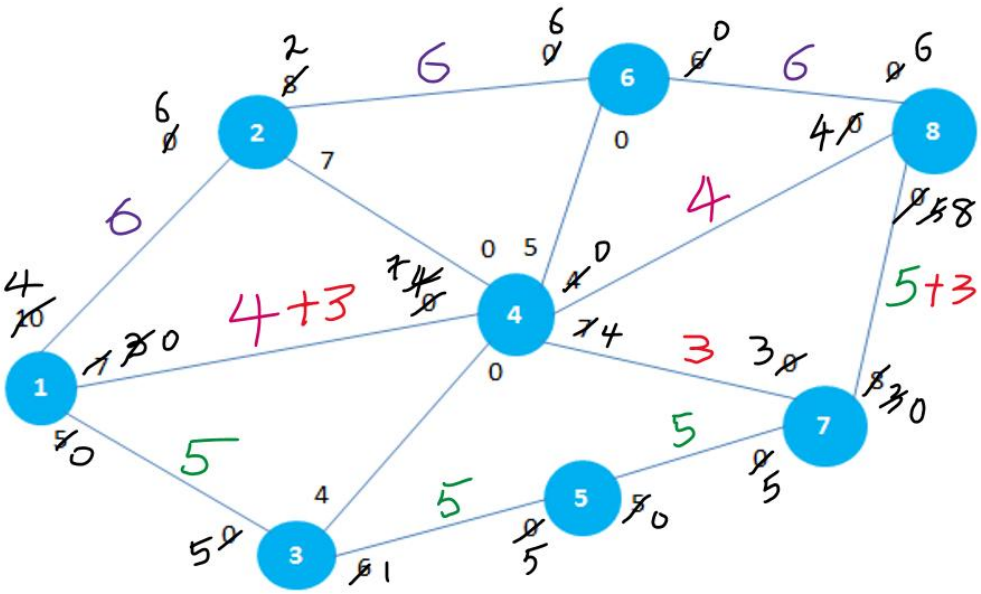
Question 2

Determine the maximum flow from source node 1 to destination node 8. Show your step-by-step workings clearly. The values on the arcs represent the capacity between 2 nodes along the direction of flow.





Path	Flow
1-2-6-8	6
1-4-8	4
1-3-5-7-8	5
1-4-7-8	3
Total Flow	18

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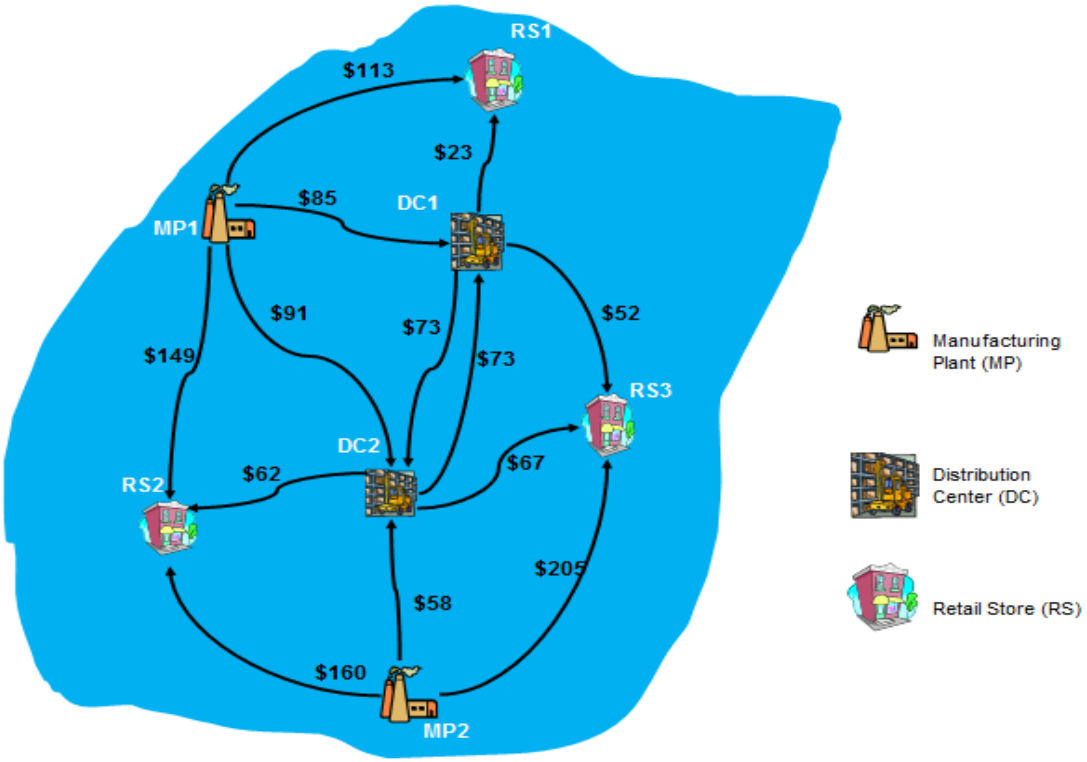
Question 3

D&T has two manufacturing plants MP1 and MP2. They ship a product to three retailers RS1, RS2 and RS3 directly or through two distribution centers DC1 and DC2. The distribution network available for shipping the product is illustrated in the following figure where the numbers on the arcs represent the unit shipping cost.



The supplies from the two plants are 400 (MP1) and 630 (MP2) units, and the demands from the three retailers are 360 (RS1), 250 (RS2) and 260 (RS3) units for the month. The distribution center 2 (DC2) also has a demand of 160 units. In addition, the shipment cannot exceed 180 units from MP2 to DC2, and 130 units from MP1 to DC1. Help D&T develop a Linear Programming (LP) model to determine an optimal distribution plan.

a) Draw a network representation of the distribution problem.

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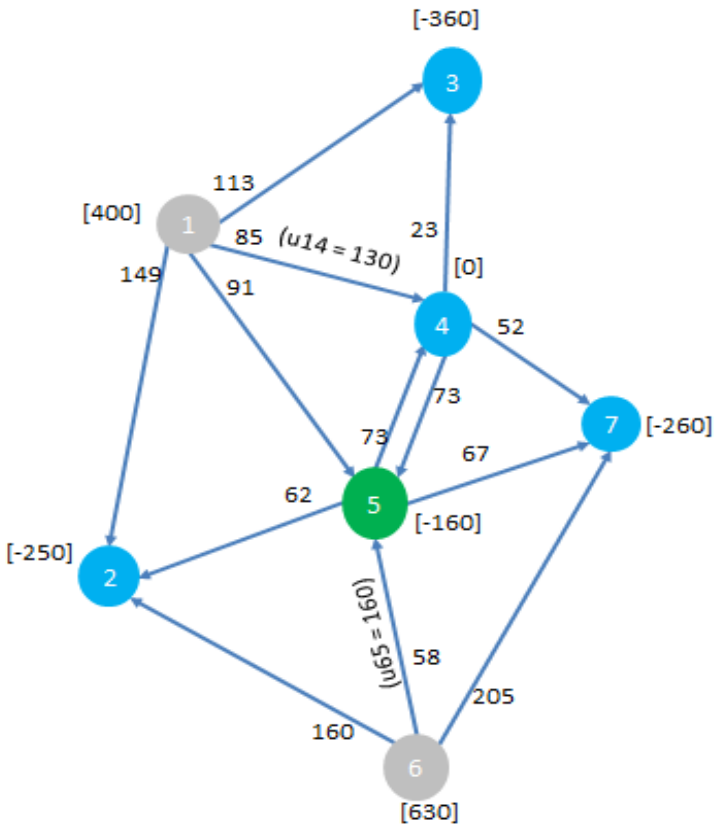

b) Define the decision variables.

Let X_{ij} represents the number of products to be shipped from node i to node j.
Where i & j = 1, 2, 3, …, 7

c) What is the objective of the LP model? Write down the objective function.

Objective is to minimize the total cost of shipping products from the manufacturing plants to the demand centers.

Objective function:
Minimize total shipping cost, $Z = 149X_{12} + 113X_{13} + 85X_{14} + 91X_{15} + 23X_{43} + 73X_{45} + 52X_{47} + 62X_{52} + 73X_{54} + 67X_{57} + 160X_{62} + 58X_{65} + 205X_{67}$

d) Formulate the constraints.

Net flow constraints

Amount shipped out – Amount shipped in = Net flow at the node

Node1: $X_{12} + X_{13} + X_{14} + X_{15} = 400$
Node2: $-X_{12} – X_{52} - X_{62} = -250$
Node3: $- X_{13} – X_{43} = -360$
Node4: $X_{43} + X_{45} + X_{47} – X_{14} – X_{54} = 0$
Node5: $X_{52} + X_{54} + X_{57} – X_{15} – X_{45} – X_{65} = -160$
Node6: $X_{62} + X_{65} + X_{67} = 630$
Node7: $-X_{47} – X_{57} – X_{67} = -260$

*Since total supply = total demand, the '=' sign is used in the net flow constraints.

If **total supply > total demand**, change to '<=' sign for all supply nodes.
E.g. Node1: $X_{12} + X_{13} + X_{14} + X_{15} <= 400$

If **total demand > total supply**, change '>=' sign for all demand nodes.
E.g. Node7: $-X_{47} – X_{57} – X_{67} >= -260$

Upper bound constraints

Flow through an arc cannot exceed the arc capacity

$X_{14} <= 130$
$X_{65} <= 160$

Non-negativity constraints
All $X_{ij} >= 0$

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Question 4

On a certain day, you need to formulate a plan to determine the sequence to deliver a certain product (packed in boxes) from warehouse (1) to six customers (2, 3, 4, 5, 6, 7). The following table shows the distance (in km) between the warehouse and each customer.

	1	2	3	4	5	6	7
1	-	26	14	30	28	29	31
2	26	-	15	23	25	8	13
3	14	15	-	14	16	15	11
4	30	23	14	-	18	21	35
5	28	25	16	18	-	33	23
6	29	8	15	21	33	-	9
7	31	13	11	35	23	9	-

a) Is this a routing or a scheduling problem? Explain.

Routing problem.

Routing involves forming the sequence in which locations are to be visited.

Scheduling involves planning the timing for each location to be visited.

b) Apply the Nearest Neighbour method to develop a delivery route.

1 - 3 - 7 - 6 - 2 - 4 - 5 - 1

c) Apply the Clarke & Wright Savings method to plan a delivery route. The savings table are derived:

Sij	1	2	3	4	5	6	7
1	0	0	0	0	0	0	0
2	0	0	25	33	29	47	44
3	0	25	0	30	26	28	34
4	0	33	30	0	40	38	26
5	0	29	26	40	0	24	36
6	0	47	28	38	24	0	51
7	0	44	34	26	36	51	0

i. Verify the Savings value for the 3-4 pair and 2-7 pair.

$S_{ij} = c_{1i} + c_{1j} - c_{ij}$ for i and j = nodes 2, 3, ..., n.

Savings 3-4 = $C_{13} + C_{14} - C_{34}$ = 14 + 30 – 14 = 30 km

Savings 2-7 = $C_{12} + C_{17} – C_{27}$ = 26 + 31 – 13 = 44 km

ii. Using the ranked Savings table given, construct the delivery route, showing clearly the sub-tour(s) formed at each step.

Rank	Node Pair	Savings
1	6--7	51
2	2--6	47
3	2--7	44
4	4--5	40
5	4--6	38
6	5--7	36
7	3--7	34
8	2--4	33
9	3--4	30
10	2--5	29
11	3--6	28
12	3--5	26
13	4--7	26
14	2--3	25
15	5--6	24

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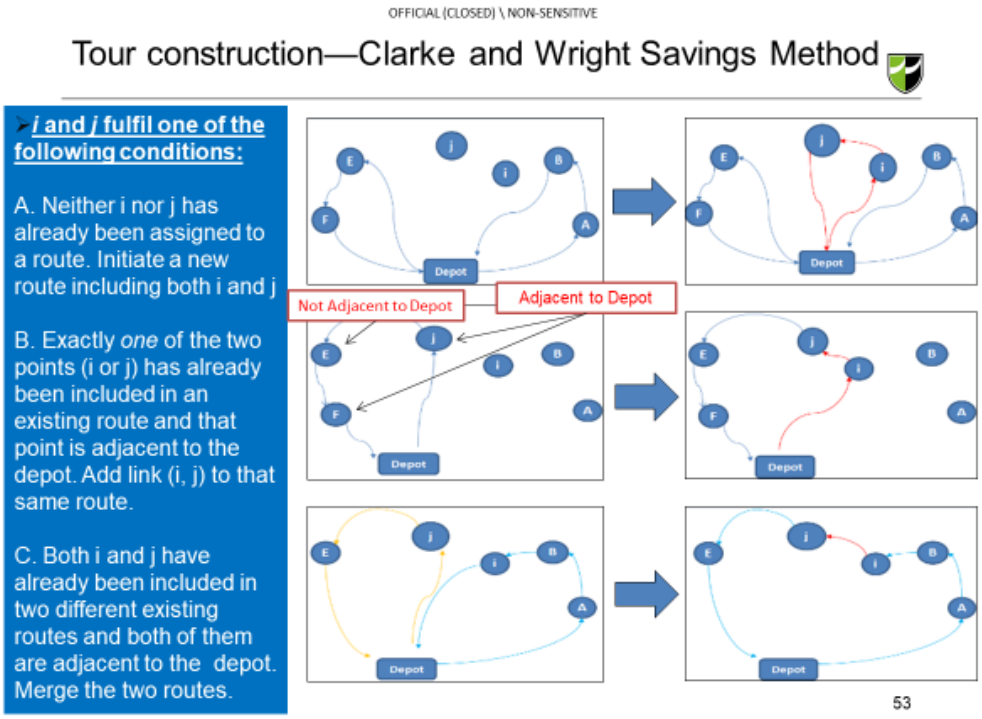

Rank	Path	Remark
1	1-6-7-1	Condition A
2	1-2-6-7-1	Condition B
4	1-2-6-7-1, 1-4-5-1	Condition A
6	1-2-6-7-5-4-1	Condition C
9	1-2-6-7-5-4-3-1	Condition B

d) Compare the total travel distance for the routes derived from the Nearest Neighbour method and Clarke & Wright Savings method.

Distance (Nearest Neighbour) = 111 km

Distance (Clarke & Wright Savings) = 112 km

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Question 5

As an operations manager, Mr Lee needs to plan the amount of a certain product to produce according to a monthly forecast derived from the historical sales. Mr Lee has collected the following data for the past 12 months:

Month	Sales for the product (kg)
1	1025
2	1250
3	1115
4	955
5	1200
6	1350
7	1025
8	1250
9	1150
10	1350
11	1050
12	1150

Show your workings clearly and round off your answers to **two** decimal places for the following questions.

a) Calculate the forecast for month 13 based on three-month and five-month moving average. Do you choose a three-month moving average or a five-month moving average if you want your forecast to be **less responsive** to the most recent data?

F_{13} (Three-month moving average) = (1150 + 1050 + 1350)/ 3
= 1183.33

F_{13} (Five-month moving average) = (1150 + 1050 + 1350 + 1150 + 1250)/ 5
= 1190.00

Choose the **five-month moving average** (higher N) forecast for the forecast to be less responsive to the most recent data.

b) Given the weights of 8, 5 and 2, calculate the three-month weighted moving average forecast for month 13.

F_{13} (Three-month weighted moving average)
= (1150 x 8 + 1050 x 5 + 1350 x 2)/ (8+5+2)
= 1143.33

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c) Given the simple exponential smoothing forecast for months 9 through 12, using Alpha (α) of 0.2 and 0.4, compute:

Month	Sales for the product (kg)	Exponential Smoothing (Alpha = 0.2)	Exponential Smoothing (Alpha = 0.4)
9	1150	1141.48	1180.29
10	1350	1143.18	1168.17
11	1050	1184.55	1240.90
12	1150	1157.64	1164.54
13		????	?????

i. Forecast for month 13 using simple exponential smoothing with alpha (α) of 0.2

F_{13} (Simple exponential smoothing α=0.2)
= 1150 x 0.2 + 1157.64 x (1-0.2) = 1156.11

ii. Forecast for month 13 using simple exponential smoothing with alpha (α) of 0.4

F_{13} (Simple exponential smoothing α=0.4)
= 1150 x 0.4 + 1164.54 x (1-0.4) = 1158.72

iii. Which forecast is less responsive to the most recent data? Explain.

The forecast based on alpha = 0.2 is less responsive to the most recent data. Smaller smoothing constant means less weight on the recent data.

d) Mr Lee is told that for the next 12 months, the sales of the product are likely to decrease. Should any of the methods above (Moving Average, Weighted Moving Average or Simple Exponential Smoothing) be used to forecast the sales?

None of the methods should be used because their forecasts will lag the trend (decreasing sales) in the data. The forecast method that he should consider is Holt's method. He can also use qualitative forecasting methods such as grassroots forecasting.

e) Calculate the Absolute Percent Error for Month 11 in the table given.

Month	Sales for the product (kg)	Forecast by Exponential Smoothing (kg)
7	1025	1136.69
8	1250	1114.35
9	1150	1141.48
10	1350	1143.18
11	1050	1184.55
12	1150	1157.64

Absolute Percent Error = abs(difference between actual and forecast)/(actual) *100%

= |1050-1184.55| / 1050 * 100% = 12.81%

<u>Question 6</u>

A 2nd hand car sales manager wants to study the correlation between car mileage and resale price. Sales price of 12 cars are plotted against their mileage. A regression line is then plotted.

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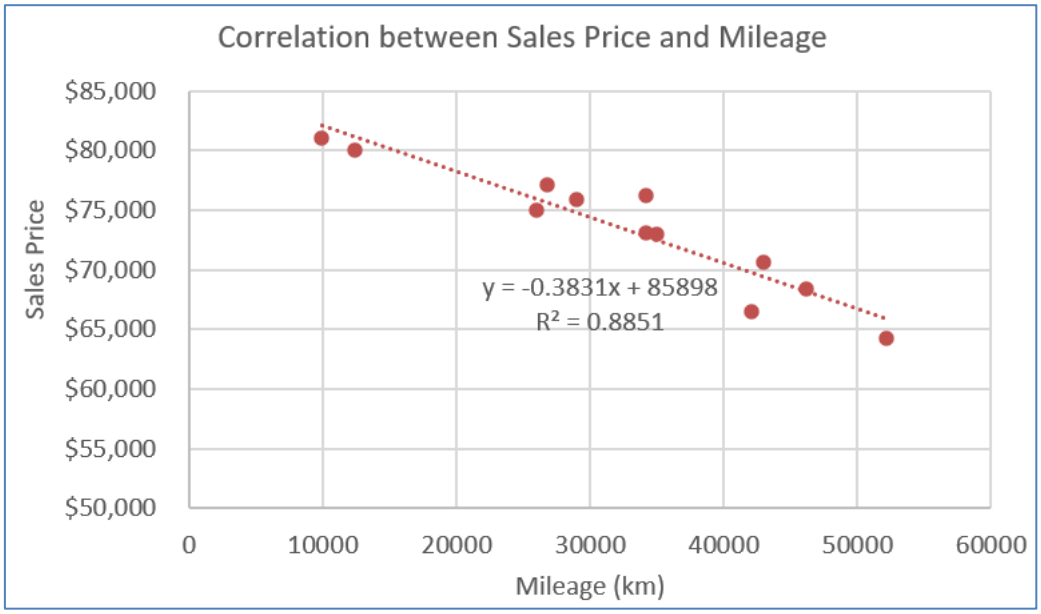

a. Identify the dependent variable and the independent variable in the regression model.

Dependent variable – car sales price

Independent variable – car mileage

b. Calculate the correlation coefficient R. Comment on the correlation between car sales price and mileage.

R = - √0.8851 = -0.941 (negative correlation)

There is a strong negative correlation between car sales price and mileage.

c. Predict the sales price of a 2nd hand car with mileage of 32000km.

Sales price, y = -0.3831x + 85898
= -0.3831*(32000) + 85898
= $73638.8

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Question 7

Amy is using the multiplicative decomposition method to forecast based on weekly sales data from the months September to November. The following table shows the Excel template she used to do her forecast.

Show your workings clearly and round off your answers to **two** decimal places for the following questions.

Week	Actual value	Time period	MA	CMA	Seasonal ratio	Seasonal Index	Deseasonalized value
SEPW1	256	1				0.86	299.28
SEPW2	322	2	381.00			0.90	356.92
SEPW3	368	3	415.75	398.38	0.92	0.94	391.15
SEPW4	578	4	A	433.00	1.33	1.30	444.06
OCTW1	395	5	492.00	471.13	0.84	0.86	461.78
OCTW2	460	6	536.25	B	0.89	0.90	509.88
OCTW3	535	7	586.75	561.50	0.95	0.94	568.65
OCTW4	755	8	634.00	610.38	1.24	1.30	580.05
NOVW1	597	9	675.25	654.63	C	0.86	697.93
NOVW2	649	10	721.50	698.38	0.93	0.90	719.37
NOVW3	700	11				0.94	D
NOVW4	940	12				1.30	722.18
						intercept	265.90
						slope	42.37

a. State the number of averaging periods used to calculate the MA.

Averaging period for MA calculation is 4, because there are 4 periods in one cycle – 4 weeks (seasons) in one month.

b. Calculate the missing entries illustrated as A, B, C and D in the Excel template.

A:
MA = (368 + 578 + 395 + 460)/ 4 = 450.25
Averaging period for MA calculation is 4, because there are 4 periods (weeks) in each repeating cycle (month).

B:
CMA = (492 + 536.25)/ 2 = 514.125
Averaging period for CMA calculation is 2.

C:
Seasonal ratio = Actual demand / CMA of that period
Seasonal Ratio = 597/654.63 = 0.912

D:
Deseasonalized Value = Actual Demand / Seasonal Index of that period
Deseasonalized Value = 700/0.94 = 744.68

c. Calculate the seasonalized forecast for Week 2 of December:

i. What is the linear equation used to forecast the deseasonalized values?

Y = 265.9 + 42.37X

ii. Calculate the deseasonalized forecast for Week 2 of December.

Index period for Week 2 of December is 14.

Deseasonalized forecast for Week 2 of December,

Y = 265.9 + 42.37* 14 = 859.08

iii. Calculate the seasonalized forecast for Week 2 of December.

Seasonalized forecast for Week 2 of December,

F_{14} = 859.08 * 0.90 = 773.17

Question 8

GoodHealth is a clinic with one duty doctor. On average, the doctor needs to spend about 5 minutes on each patient. The average number of patients arriving at the clinic every hour is 10.

Show your workings clearly and round off your answers to **two** decimal places for the following questions.

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a. State the queuing model you can use to calculate the queue performance measures at the clinic. What are the assumptions needed?

M/M/1

- Input source: Infinite; no balks, no reneging
- Arrival process: Number of arrivals follows Poisson distribution
- Queue: Unlimited; single line
- Queue discipline: FIFO (FCFS)
- Service time: Negative exponential distribution
- Service rate is independent of arrival rate

b. Calculate the mean arrival rate and mean service rate in per minute.

Mean arrival rate, λ: 10/60 = 1/6 patient per minute

Mean service rate, μ: 1/5 = 0.2 patient per minute

c. Calculate the utilization, ρ.

Utilization, ρ = arrival rate/ service rate = (1/6)/ (1/5) = 0.83

d. Calculate the average number of patients waiting for the doctor, Lq.

Avg number waiting, Lq = $\frac{\rho^2}{1-\rho}$ = 0.83^2/ (1-0.83) = 4.05

e. Calculate the average time a patient spends waiting for the doctor, Wq.

Avg waiting time time, Wq = Lq/ λ = 4.05/ (1/6) = 24.3 minutes

f. Calculate the average time a patient spends at the clinic, Ws.

Ws = Wq + 1/μ [1/μ is service time]
= 24.3 + 5
= 29.3 minutes

g. Suppose the patient arrival rate increases to 23 per hour during the flu season. What is the minimum number of doctors required for the queueing system to be in a steady state condition? Average service time remains at 5 minutes.

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Steady state condition:
Effective arrival rate / Effective service rate <1
Arrival rate/ (k x service rate) < 1

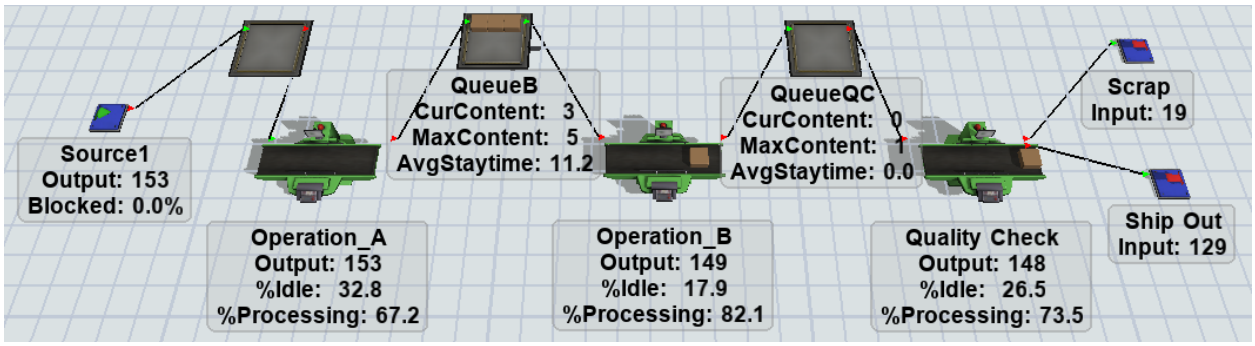
where k is the number of servers (doctors)
(23/60)/ (k x 0.2) < 1
k > (23/60)/ 0.2 > 1.92 ~ 2 (minimum number of k needed)

Arrival rate = 23/60 = 0.38 per min
Service rate = 1/5 = 0.2 per min
Steady state equilibrium holds when arrival rate < k(service rate) [0.38 < k(0.2)]
Minimum k > arrival rate/service rate > 0.38/0.2 > 1.9
Minimum number of workers required, k =2

Question 9

A production line has three major operations: Operation_A, Operation_B, and Quality Check.

a. A simulation model is built for the production line. A snapshot of the simulation as shown.



State the following statistics:

i) Number of items processed by Operation_A = 153

ii) Number of items that are shipped out = 129

iii) Average waiting time at Operation_B = 11.2

iv) Maximum queue length at Operation_B = 5

v) Utilization of Quality Check = 73.5%

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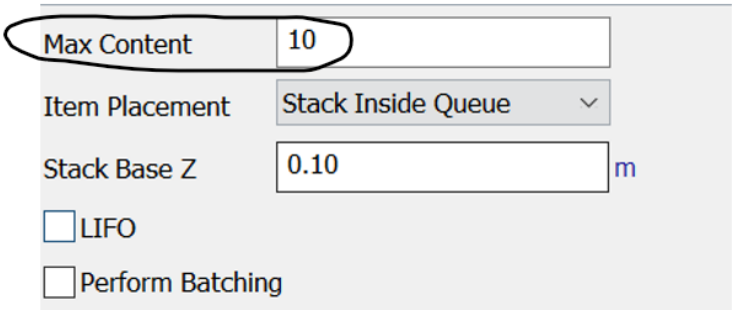
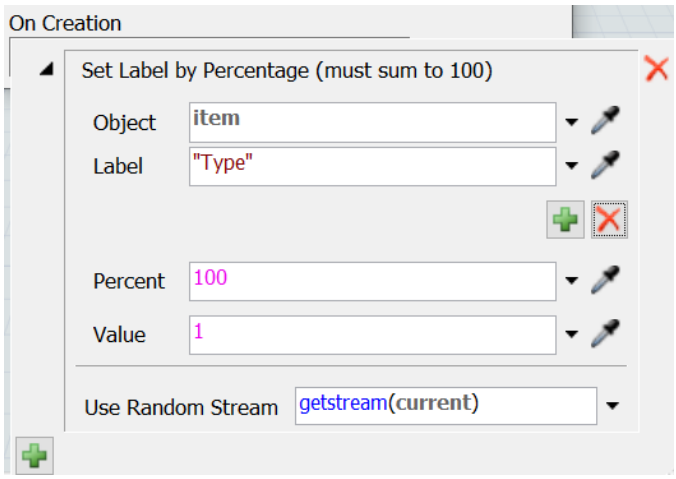
b. Suppose there is limited space for the work-in-progress parts in the production line. How do you model this?

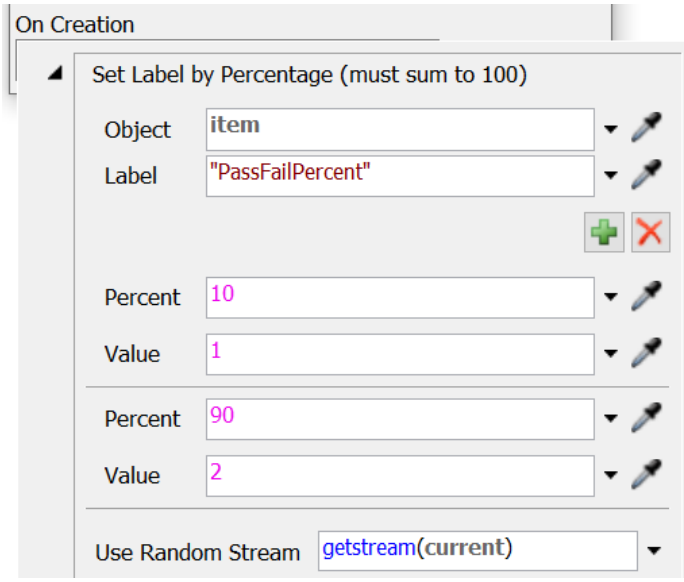
Set the max content of the queues to reflect the amount of space available for queuing.



c. After the products go through Quality Check, they are either sent to Scrap (failed QC) or Ship Out (passed QC). The percentage of products that passes QC is 90%. To model this, a label named 'PassFailPercent' is to be created for each item. Items with label value of 1 will be sent to Scrap whereas those with label value of 2 will be sent to Ship Out. How do you set this at the On Creation trigger of Source1?



First, change Label name to "PassFailPercent".
Then, add a case.
Finally, set the Percent and Value of each case as shown below.



Question 10

A process flow simulation model was built for a warehouse picking and packing process as shown in the figure below.

a. State the number of pickers in the process and the number of items waiting to be picked.

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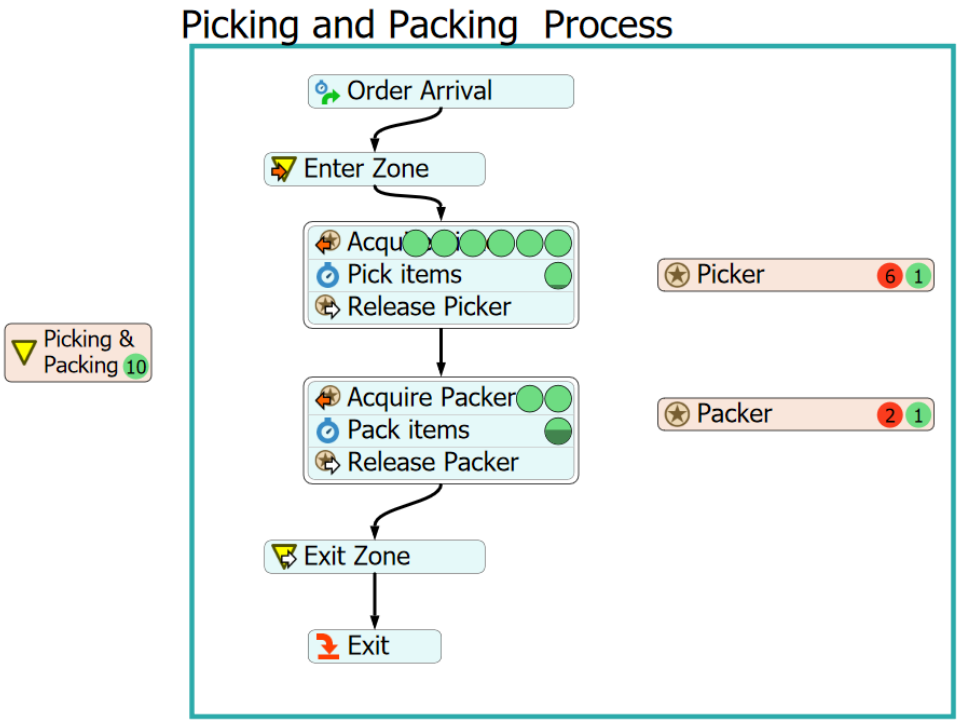


Number of pickers = 1

Number of items waiting to be picked = 6

b. What is the purpose of the activities 'Acquire Packer' & 'Release Packer'?

'Acquire Packer' is to obtain an available Packer resource before Pack items can be carried out.

'Release Packer' is to return the Packer resource to the available state for the next token (item) to acquire.

c. Figure below shows the statistics of the Picker.

i) What is the Average and Max number of orders waiting for the Picker?

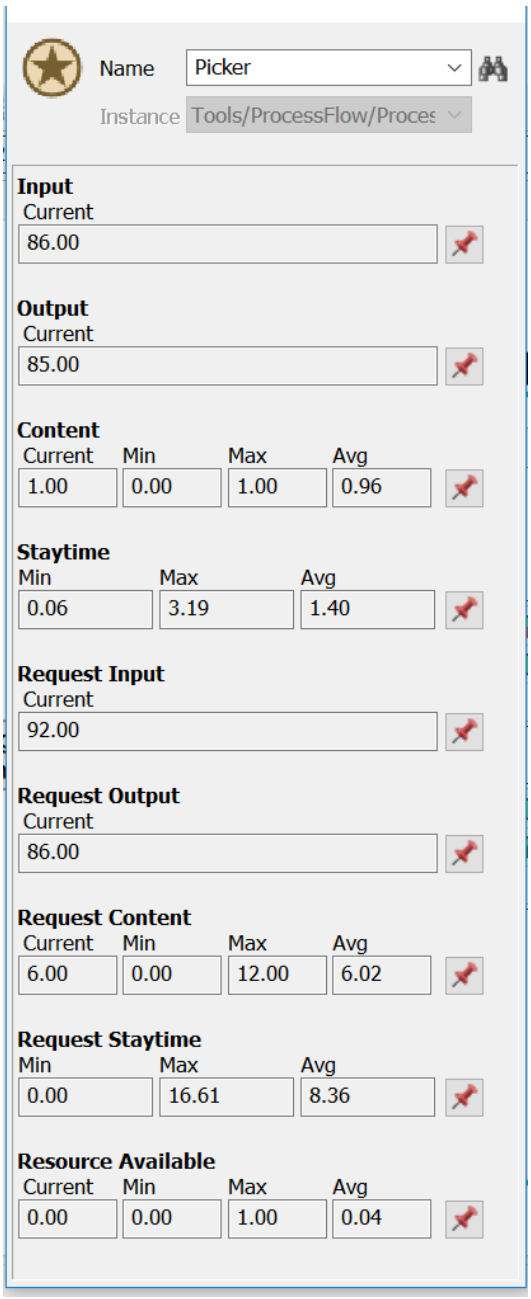
"Request Content"
Average: 6.02, Max: 12

ii) What is the Average and Max waiting time for the Picker?

"Request Staytime"
Average: 8.36 mins, Max: 16.61 mins

iii) What is the average utilization of the Picker?

Content (Avg) / Resouce Available (Max) = 0.96/1 *100% = 96%



d. How would you know who is the bottleneck, Picker or Packer?

Compare the waiting time, queue length or utilization of the Picker and Packer. The resource with the higher waiting time/ queue length/ utilization is the bottleneck.